

Agile Software Development and Its Core Principles

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Introduction to Agile Software Development

- Agile Software Development is a methodology that promotes continuous iteration of development and testing throughout the software development lifecycle. It emphasizes flexibility, collaboration, and customer feedback to deliver high-quality software efficiently.

Key Features of Agile

- - Iterative and incremental development
- - Adaptive planning and evolutionary design
- - Early delivery and continuous improvement
- - Close collaboration between cross-functional teams
- - Focus on customer satisfaction

Popular Agile Methodologies

- 1. Scrum – Based on sprints and daily stand-up meetings.
- 2. Kanban – Uses a visual board to track workflow and limit work in progress.
- 3. Extreme Programming (XP) – Focuses on engineering excellence and frequent releases.
- 4. Lean Software Development – Reduces waste and maximizes value.
- 5. Crystal – Adapts process depending on project size and criticality.

Agile Manifesto

- The Agile Manifesto outlines four core values:
- 1. Individuals and interactions over processes and tools.
- 2. Working software over comprehensive documentation.
- 3. Customer collaboration over contract negotiation.
- 4. Responding to change over following a plan.

Core Principles of Agile

- 1. Customer satisfaction through early and continuous delivery.
- 2. Welcome changing requirements, even late in development.
- 3. Deliver working software frequently.
- 4. Collaboration between business people and developers.
- 5. Build projects around motivated individuals.
- 6. Simplicity—the art of maximizing the amount of work not done—is essential.

Benefits and Conclusion

- Benefits of Agile include:
 - - Faster time to market.
 - - Improved product quality.
 - - Greater customer satisfaction.
 - - Better team morale and collaboration.
- Conclusion:
 - Agile Software Development is a powerful and flexible approach that helps teams deliver high quality software efficiently by embracing